URBANROOF: DETAILED DIAGNOSTIC REPORT

STATUS: HIGH SEVERITY - MOISTURE INGRESS DETECTED

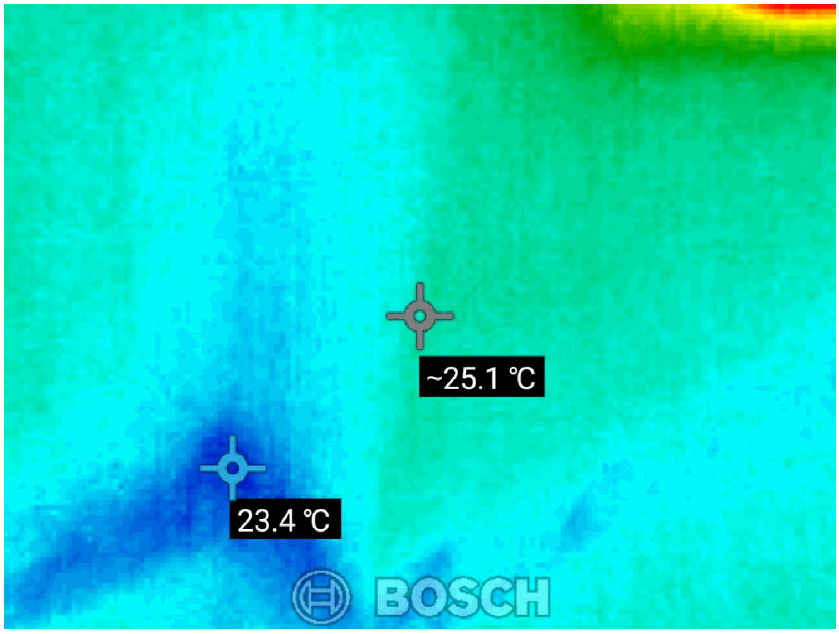
Detailed Diagnostic Report (DDR)

Date of Report: [Current Date]
Inspection Date: 27.09.2022
Inspected By: Krushna & Mahesh

Property Information:
* Customer Name: Not Available
* Mobile: Not Available
* Email: Not Available
* Address: Not Available
* Property Age: Not Available
* Property Type: Flat
* Floors: 11
* Previous Structural Audit: No
* Previous Repair Work: No

1. Property Issue Summary

The inspection revealed widespread moisture-related issues, primarily dampness and seepage, affecting multiple internal areas including the hall, bedrooms, kitchen, and common bathroom ceiling. External wall cracks and potential plumbing problems both within the unit and potentially from an adjacent unit appear to be contributing factors. Tile hollowness and joint issues were also noted in bathrooms. Thermal imaging strongly supports the presence of moisture in affected areas.



Thermal Analysis Reference: Hall

Thermal Analysis Reference: Bedroom

Thermal Analysis Reference: Kitchen

Thermal Analysis Reference: Bathroom

2. Area-wise Observations

Hall:

* Dampness: Observed at the skirting level.
* Thermal Validation: Cold spots observed at wall bases and skirting levels indicate moisture presence, aligning with the dampness finding.

Bedroom (Common):

* Dampness: Observed at the skirting level.
* Bathroom Tiles: Tile hollowness observed in the common bathroom.
* Thermal Validation: Cold spots observed at wall bases and skirting levels indicate moisture presence, aligning with the dampness finding.

Master Bedroom:

* Dampness: Observed at the skirting level and on wall surfaces (with efflorescence).
* Bathroom Tiles: Tile hollowness observed in the master bedroom bathroom.
* Thermal Validation: Cold spots observed at wall bases and skirting levels indicate moisture presence, aligning with the dampness finding.

Kitchen:

* Dampness: Observed at the skirting level.
* Thermal Validation: Cold spots observed at wall bases and skirting levels indicate moisture presence, aligning with the dampness finding.

Common Bathroom (General):

* Tile Condition: Tile hollowness is widespread. Gaps/blackish dirt observed in tile joints and around Nahani Trap joints.
* Plumbing: Loose plumbing joints/rust observed around flush tank, shower, angle cock, bibcock, and washbasin. Leakage due to concealed plumbing and damage in Nahani trap/brickbat coba under tile flooring. Plumbing issue noted.
* Ceiling: Mild dampness observed on the ceiling.
* Leakage: Leakage is observed to be continuous (all time).
* Thermal Validation: Cold spots indicating moisture are likely present, supporting the dampness and leakage findings.

Parking Area (Below Flat):

* Seepage: Observed on the ceiling of the parking area, indicating leakage from the flat above.
* Thermal Validation: Cold spots indicating moisture are likely present, supporting the seepage finding.

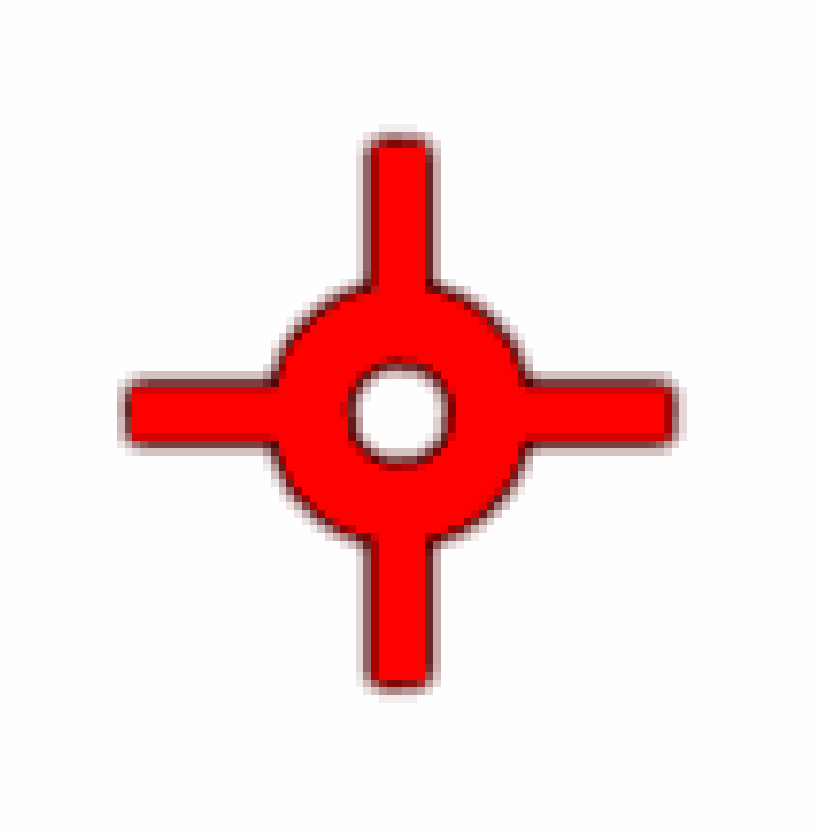
External Walls:

* Cracks: Moderate cracks observed on the external surface near the Master Bedroom and on RCC Column and Beam.
* Duct Issue: An issue related to a duct was noted.
* Plumbing Pipes: External plumbing pipes are moderately cracked and leaking.

* Vegetation: Moderate algae, fungus, and moss observed.
* Thermal Validation: While thermal imaging directly on external walls wasn't detailed, external issues can contribute to internal dampness, which is confirmed by thermal data.

Adjacent Flat (Flat No. 203):
* Tile Joints: Tile joints are open, and an outlet leakage issue was noted. This could be a source of leakage affecting the Common Bathroom ceiling of Flat No. 103.

No conflicting observations were found within the provided reports.



Thermal Analysis Reference: Master Bedroom

Thermal Analysis Reference: Parking

3. Probable Root Cause

The widespread dampness at skirting levels, wall surfaces, and ceilings strongly suggests multiple sources of moisture ingress. Probable root causes include:

* Internal Plumbing Leaks: Identified issues with concealed plumbing, Nahani trap/brickbat coba damage, and loose plumbing joints in bathrooms are primary candidates for internal dampness and seepage.
* External Wall Issues: Moderate cracks on external walls, cracked external plumbing pipes, and algae/moss growth can allow rainwater ingress, contributing to internal dampness, especially near the master bedroom.
* Bathroom Tile and Joint Deterioration: Tile hollowness, open tile joints, and gaps around Nahani traps can compromise waterproofing, allowing water to seep into the slab and walls.
* Leakage from Upper Floors/Adjacent Units: The noted tile joint issues and outlet leakage in Flat No. 203 strongly indicate a potential source of leakage affecting the Common Bathroom ceiling of Flat No. 103.

4. Severity Assessment

Severity: Significant

Reasoning:
* Widespread Impact: Dampness is present in multiple critical areas (hall, bedrooms, kitchen, common bathroom,

parking ceiling).

* Continuous Leakage: Leakage in the WC and external wall areas is noted as "All time," indicating an ongoing issue rather than intermittent.
* Structural Concerns (Moderate): Moderate cracks in RCC columns/beams and external walls, along with leaking external plumbing, suggest potential for structural deterioration if not addressed.
* Thermal Confirmation: The consistent cold spots indicating moisture presence across affected areas validate the severity of the dampness observed visually.
* Potential for secondary damage: Prolonged dampness can lead to mold growth, degradation of finishes, and compromise structural integrity over time.

5. Recommended Actions

1. Comprehensive Plumbing Inspection & Repair:
   * Thoroughly inspect all internal plumbing lines in the common and master bathrooms, including concealed pipes, Nahani traps, and brickbat coba.
   * Repair or replace all identified leaking or loose plumbing joints (flush tank, shower, angle cock, bibcock, washbasin).
   * Address the concealed plumbing leakage and Nahani trap/brickbat coba damage.
2. Bathroom Waterproofing & Tile Work:
   * Remove hollow tiles in both common and master bathrooms.
   * Re-grout all open or damaged tile joints, including those around Nahani traps, ensuring a proper waterproof seal.
   * Consider re-waterproofing the bathroom floor areas, especially if brickbat coba damage is significant.
3. External Wall Repairs:
   * Repair all moderate cracks observed on the external walls and RCC columns/beams.
   * Address the "Duct Issue" on the external wall.
   * Repair or replace cracked and leaking external plumbing pipes, ensuring proper sealing around pipe openings.
   * Clean off algae, fungus, and moss from external walls and apply appropriate anti-fungal treatments and paint.
4. Addressing Leakage from Flat No. 203 (if applicable):
   * Coordinate with Flat No. 203 to address their open tile joints and outlet leakage, as this directly impacts the common bathroom ceiling of Flat No. 103.
5. Internal Dampness Remediation:
   * Once the sources of leakage are fixed, allow affected wall and ceiling areas to dry completely.
   * Remove damaged plaster and re-plaster/repaint affected areas in the Hall, Bedrooms, Master Bedroom, Kitchen, and Common Bathroom ceiling.
6. Parking Area Seepage:
   * The source of the parking area seepage must be located and rectified, likely originating from the common bathroom or external areas of Flat No. 103.

6. Additional Notes

* Thermal analysis consistently identified cold spots at wall bases, corners, and skirting levels, providing strong

evidence of moisture presence and validating the observed dampness issues.

* The report indicates an overall inspection score of 85.71% and 1 flagged item, which is a high-level summary. The detailed findings, however, reveal significant and widespread issues requiring prompt attention.

7. Missing or Unclear Information

* Customer Contact Details: Mobile number and email address are "Not Available."
* Property Age: The age of the property in years is "Not Available."
* Specific Location of Duct Issue: While a "Duct Issue" is mentioned, its precise nature and impact are not detailed.
* Details of "N/A" Checklist Items: Several checklist items (e.g., sealants on window frames, AC frame condition, grout around pipes, patchwork plaster, expansion joints, corrosion on metal rods, chalking/flaking, overhead water tank leakage) are marked "N/A," which means these aspects were not assessed or were not applicable. A complete assessment might benefit from clarification on these points if relevant.
* Flat No. 203 Impact: While Flat No. 203's issues are noted as a potential source, the extent of its contribution to Flat No. 103's problems and the responsible party for resolution are not specified.